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& Technology Peshawar

MBSD PROJECT PROPOSAL

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TITLE

Smart Rain-Proof Clothes Protection System using IoT and Sensors

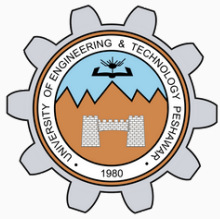
A smart automation project that uses sensors, IoT, and actuators to protect clothes from rain by retracting the clothesline when rainfall is detected. The system also provides user notifications via a buzzer and allows remote monitoring and control via the Blynk app.



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Proposal **Objective**

To design and implement an intelligent system that automatically protects hanging clothes from rainfall by retracting them inside a shelter. The system uses a rain sensor to detect precipitation, a servo motor to move the clothesline, and a buzzer to alert the user before action is taken. Additionally, a temperature and humidity sensor (SHT31) monitors the environment, and the system is integrated with the Blynk IoT app for real-time monitoring and control. Manual override is available through both a physical push button and the mobile app.



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MOTIVATION

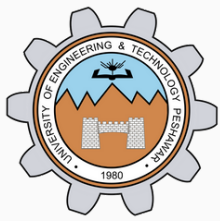
In many regions, sudden rainfall can spoil drying clothes, especially when no one is home to take immediate action. This project aims to solve that problem using an automated solution that not only protects clothes but also updates users remotely. The integration of real-time sensor data and a smart control system brings convenience, reliability, and innovation to everyday chores.



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Tools & Components

- ESP32 (Microcontroller)
- Rain Sensor Module (for rainfall detection)
- SHT31 Temperature & Humidity Sensor (for climate monitoring)
- Servo Motor (to move clothesline rod)
- Buzzer (alerts user before movement)
- Push Button (manual control)
- App (remote monitoring and control)
- Breadboard, Jumper Wires, USB Cable
- Miniature House Model and Clothesline setup (demo structure)



Working Principal

Rain Detection:

The rain sensor monitors for water droplets. If rain is detected, the buzzer rings briefly to notify the user, and after a short delay, the servo motor retracts the clothesline inside the house.

Rain Stops:

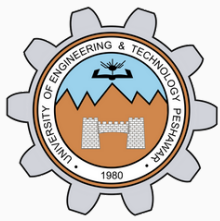
When the rain stops, the system waits for a defined period (to avoid false triggers), then extends the clothesline back outside.

Manual Control:

If automatic action fails or is not desired, the user can press a physical push button or use the Blynk app to manually retract or extend the clothes.

IoT Monitoring:

The SHT31 sensor reads temperature and humidity values continuously and sends the data to the Blynk IoT app along with the rain status.



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Conclusion

The proposed smart clothes protection system provides an effective, automated solution to shield clothes from rain while incorporating user-friendly features such as a manual override, app-based control, and real-time environmental monitoring. The addition of a buzzer enhances user interaction by providing audible alerts, making the system more practical and efficient for real-world use.